

Workshop 00: Create a Buck Converter with the Average Switch Model

Goal

Create a simple buck converter using the Simscape Electrical averaged `Buck Converter` block. This workshop follows the clip `00_Create_Buck_Simscape_Average_Simplify.mp4`.

Start from:

```
Create_Buck_Simscape_Elec__Average_Simplify.slx
```

Use the finished model as the reference:

```
Create_Buck_Simscape_Elec__Average_Simplify_finished.slx
```

The purpose of this workshop is to show a simplified buck converter workflow before using detailed switching components. The average switch model uses a duty-cycle input directly, so the converter can be simulated quickly without building the MOSFET, diode, inductor, and capacitor from separate blocks.

By the end of this workshop, participants should be able to:

- Add the Simscape Electrical `Buck Converter` block.
- Configure it for averaged switching behavior.
- Drive the converter using a constant duty-cycle command.
- Connect a resistor load and measure the output voltage.
- Add a DC motor load subsystem.
- Compare output voltage and motor speed behavior in Simulation Data Inspector.

Prerequisites

- MATLAB with Simulink.
 - Simscape and Simscape Electrical.
 - Model files provided with this workshop.
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Starting Files

File	Purpose
Create_Buck_Simscape_Elec__Average_Simplify.slx	Starting model with source, reference, duty signal, resistor load, voltage sensor, converter blocks, and scope
Create_Buck_Simscape_Elec__Average_Simplify_finished.slx	Finished reference model with the averaged buck converter and DC motor load

Starting Model

The starting model contains these main blocks:

Block	Purpose
DC Voltage Source	Input supply for the converter
Electrical Reference	Electrical ground reference
Solver Configuration	Required Simscape solver block
Constant	Duty-cycle command
Simulink-PS Converter	Converts the duty command to a physical signal
Resistor	Initial electrical load
Voltage Sensor	Measures output voltage
PS-Simulink Converter	Converts measured voltage to Simulink
Scope	Displays the measured voltage

Detected starting settings:

Item	Value
Stop time	500 s
DC source voltage	50 V
Duty command	0.5
Resistor load	1 Ohm

Finished Model Reference

The finished model adds:

Added Item	Purpose
Buck Converter	Average switch buck converter stage
DC_Motor subsystem	Motor load connected to the converter output
Scope1 inside DC_Motor	Motor speed display
Simulation Data Inspector runs	Compare output voltage and motor speed

Detected finished settings:

Item	Value
Stop time	5000 s
Buck converter device type	Averaged switching device
DC source voltage	50 V
Duty command	0.5
Resistor load	1 Ohm
Buck converter inductor Lf	1e-6 H
Buck converter capacitor Cf	1e-7 F
Switch on resistance Ron	0.001 Ohm
Switch off conductance Goff	1e-5 1/Ohm

The expected resistor-load output voltage is approximately:

$$V_{out} = \text{Duty} * V_{in} = 0.5 * 50 \text{ V} = 25 \text{ V}$$

Workshop Procedure

Follow this order in the workshop:

Open starting model -> Add averaged Buck Converter

-> Configure Buck Converter parameters -> Wire source, duty input, and resistor load

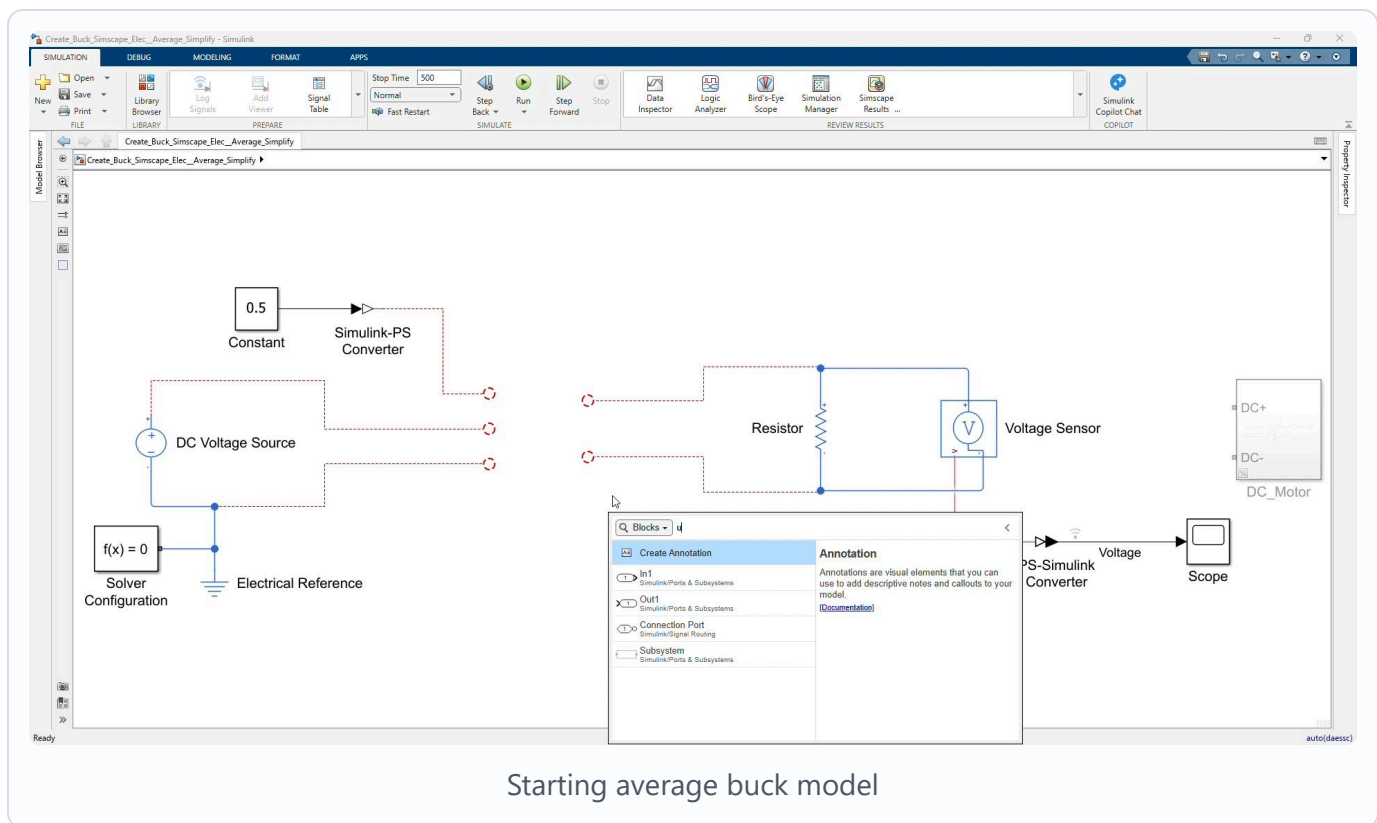
-> Run resistor-load simulation -> View output voltage

-> Create/open DC_Motor subsystem -> Configure DC motor load

-> Connect DC_Motor to converter output -> Run final simulation

-> Compare voltage and motor speed in Simulation Data Inspector

1. Open the starting average buck model



Video screenshot: starting model with the prepared source, duty command, resistor load, voltage sensor, and scope.

1. Open:

Create_Buck_Simscape_Elec_Average_Simplify.slx

1. Review the prepared blocks:

- DC Voltage Source

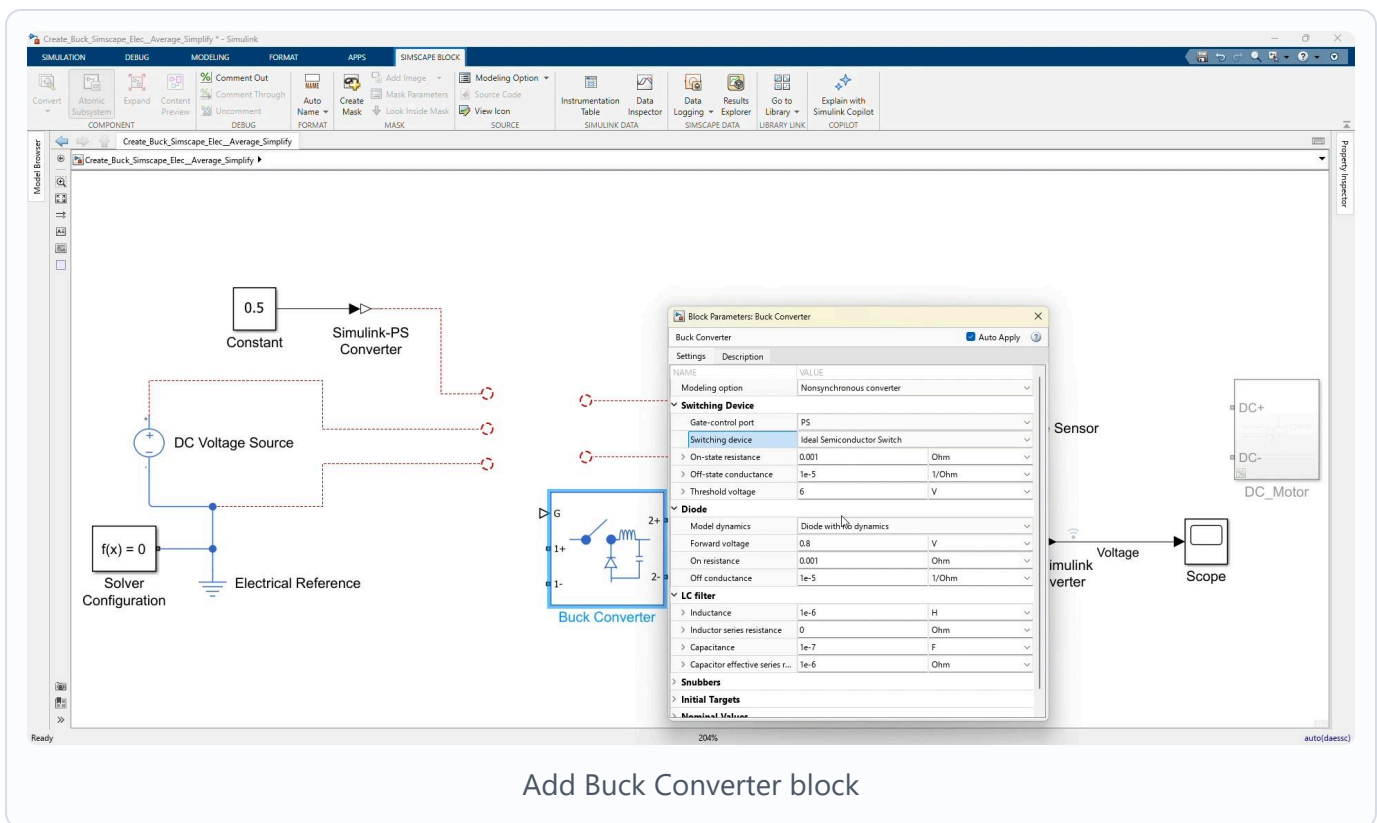
- Electrical Reference
- Solver Configuration
- Constant
- Simulink-PS Converter
- Resistor
- Voltage Sensor
- PS-Simulink Converter
- Scope

1. Confirm the duty command is:

0.5

1. Explain that this duty command will directly control the averaged buck converter.

2. Add the Buck Converter block



Video screenshot: adding the Simscape Electrical Buck Converter block to the model.

1. Open the Library Browser or use quick insert.

2. Search for:

Buck Converter

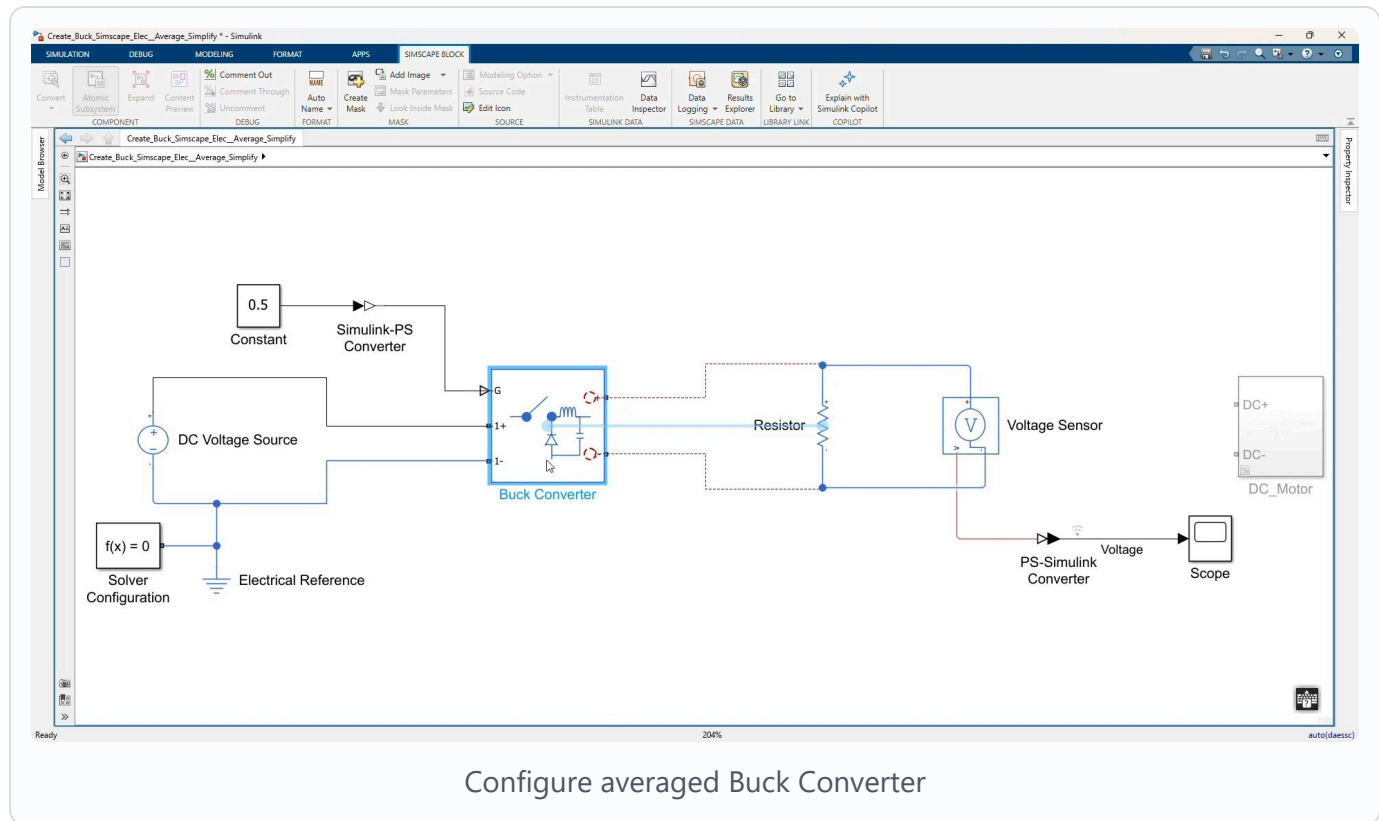
1. Add the Simscape Electrical **Buck Converter** block to the center of the model.
2. Place it between the DC input source and the output load.

Concept to explain:

The Buck Converter block represents the converter power stage as one reusable Simscape block.

For this lesson, it is configured as an averaged model, so it does not require PWM switching pulses.

3. Configure the Buck Converter as an averaged model



Video screenshot: Buck Converter block parameters configured for averaged switching behavior.

Open the **Buck Converter** block parameters.

Confirm or set the converter device type to:

Averaged switching device

Use the default/simplified converter parameters shown in the finished model:

Parameter	Value
Switch on resistance <code>Ron</code>	<code>0.001 Ohm</code>
Switch off conductance <code>Goff</code>	<code>1e-5 1/Ohm</code>
Forward voltage <code>Vf</code>	<code>0.8 V</code>
Filter inductance <code>Lf</code>	<code>1e-6 H</code>
Filter capacitance <code>Cf</code>	<code>1e-7 F</code>

The important point is not detailed semiconductor switching. The lesson is about creating a fast average converter model driven by a duty command.

4. Connect the DC source to the converter input

Connect the input side of the `Buck Converter` :

```
DC Voltage Source positive -> Buck Converter p1
DC Voltage Source negative -> Buck Converter n1
Electrical Reference -> DC Voltage Source negative / Buck Converter n1
Solver Configuration -> electrical reference node
```

The input supply is:

`50 V`

5. Connect the duty command to the converter gate input

Connect the control signal:

Constant(0.5) -> Simulink-PS Converter -> Buck Converter G

Explain:

The averaged Buck Converter block uses this physical-signal input as the duty ratio.

Duty = 0.5 means the ideal average output should be about half of the input voltage.

6. Connect the resistor load and voltage sensor

Connect the output side of the `Buck Converter`:

Buck Converter p2 -> Resistor positive

Buck Converter n2 -> Resistor negative

Voltage Sensor across the resistor

Voltage Sensor output -> PS-Simulink Converter -> Scope

Keep the resistor value:

1 Ohm

Name or identify the measured signal as:

Voltage

7. Run the resistor-load simulation

1. Run the model with the resistor load connected.
2. Open the Scope or Simulation Data Inspector.
3. Confirm the output voltage rises near:

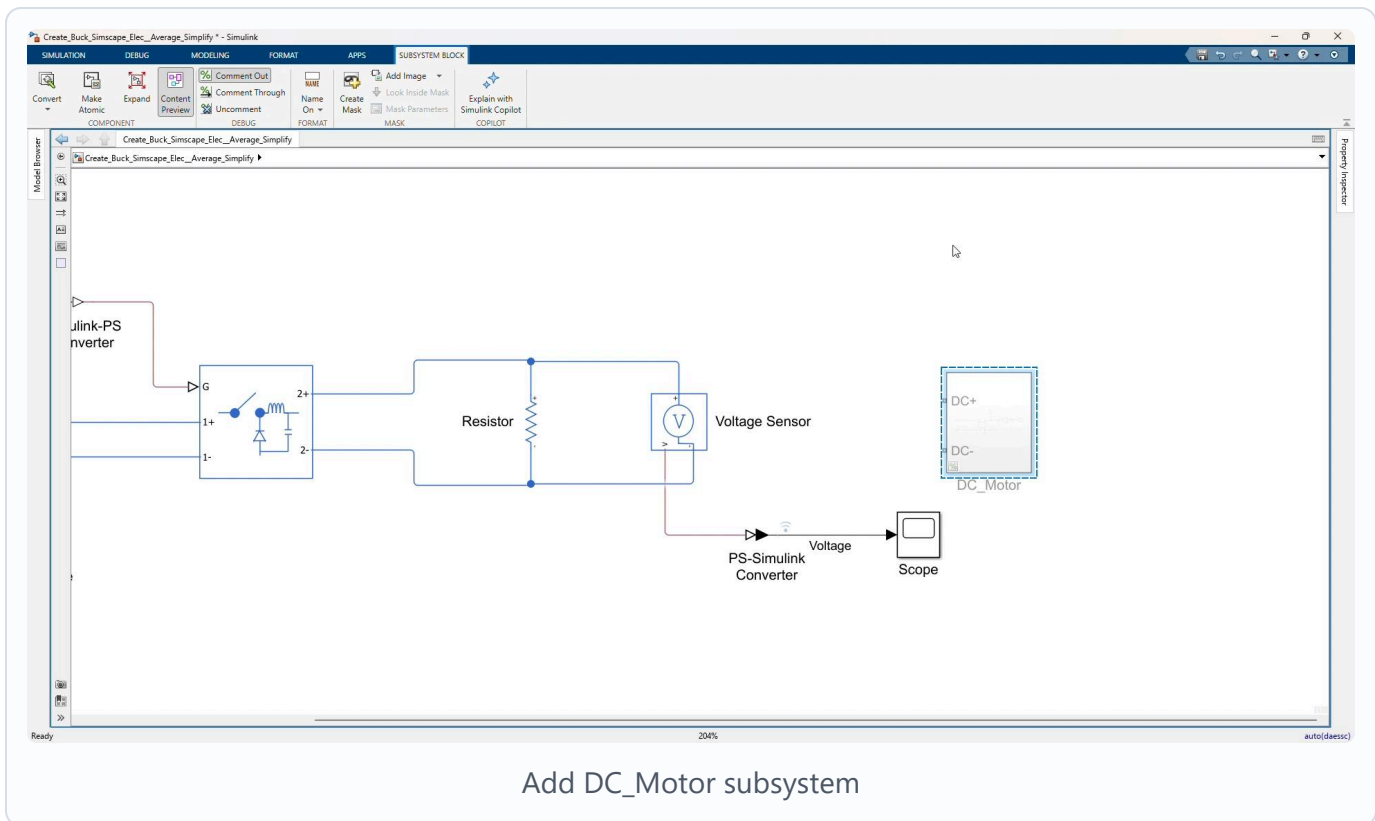
25 V

1. Explain the relationship:

$$50 \text{ V input} * 0.5 \text{ duty} = \text{about } 25 \text{ V output}$$

This validates the basic averaged buck converter connection before adding the motor load.

8. Create the DC_Motor subsystem



Video screenshot: DC_Motor subsystem added at the converter output side.

Create or open the `DC_Motor` subsystem used in the finished model.

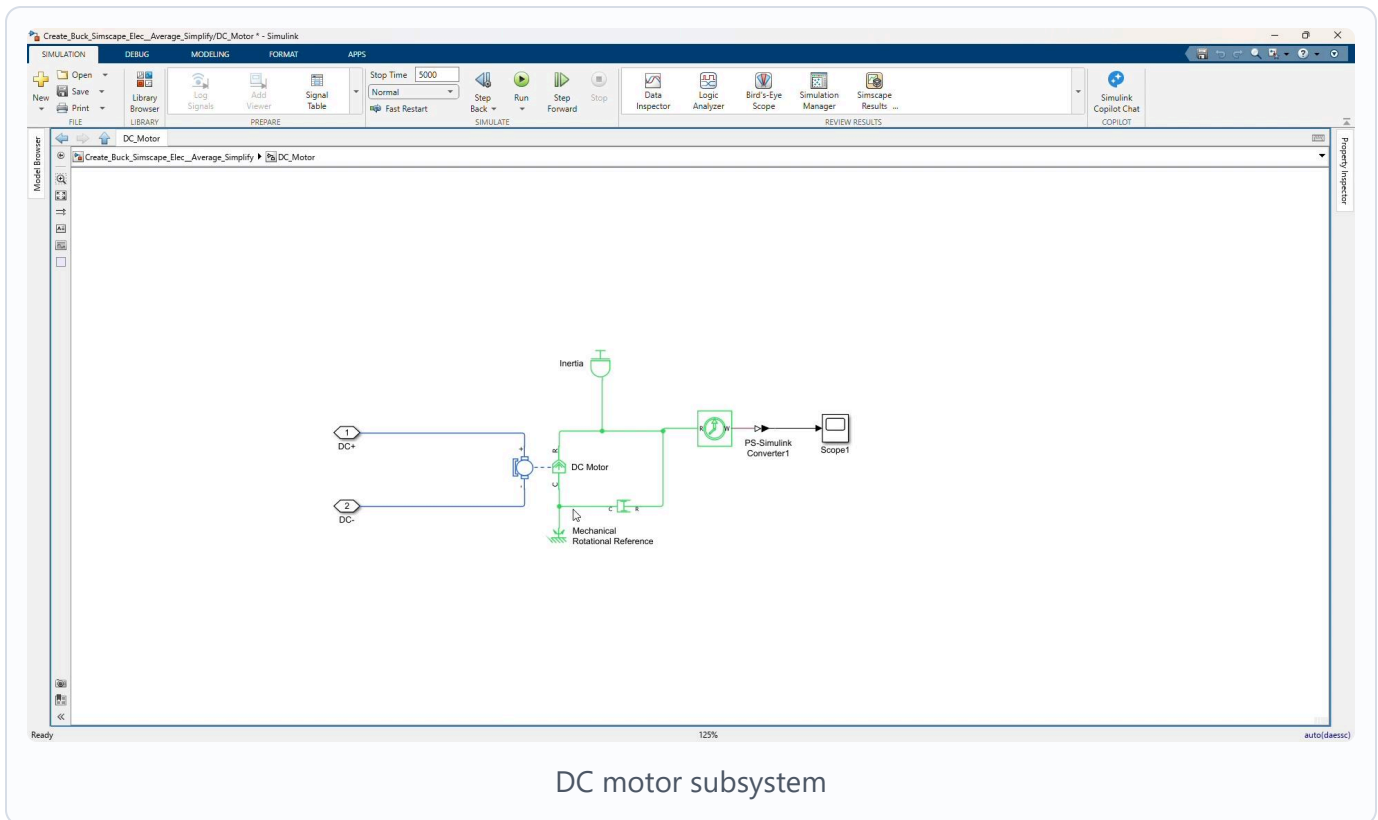
The subsystem has two electrical conserving ports:

DC+
DC-

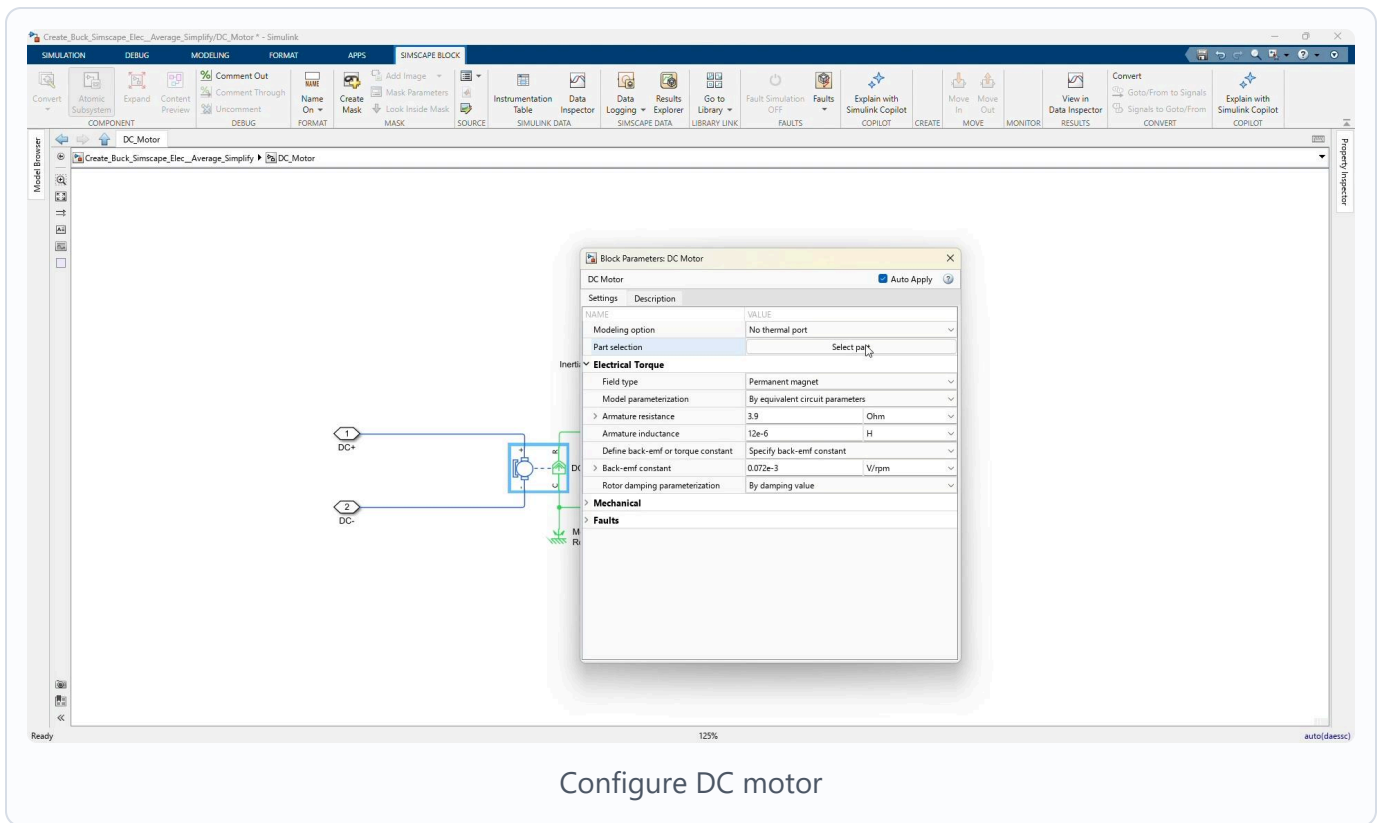
Inside the subsystem, add or review these blocks:

Block	Purpose
DC Motor	Electrical-to-rotational motor model
Inertia	Rotational load inertia
Rotational Damper	Mechanical damping load
Mechanical Rotational Reference	Mechanical reference
Ideal Rotational Motion Sensor	Measures motor speed
PS-Simulink Converter1	Converts speed to Simulink
Scope1	Displays motor speed

9. Configure the DC motor load



Video screenshot: internal DC_Motor subsystem structure with motor, inertia, damper, speed sensor, and scope.



Video screenshot: DC Motor block parameter dialog.

Use the DC motor settings shown in the finished model:

Parameter	Value
Parameterization	Circuit parameters
Armature resistance R_a	3.9 Ohm
Armature inductance L_a	12e-6 H
Back-emf constant K_v	0.072e-3 V/rpm
Rated voltage	1.5 V
Rated speed	15000 rpm
Rated power	0.08 W

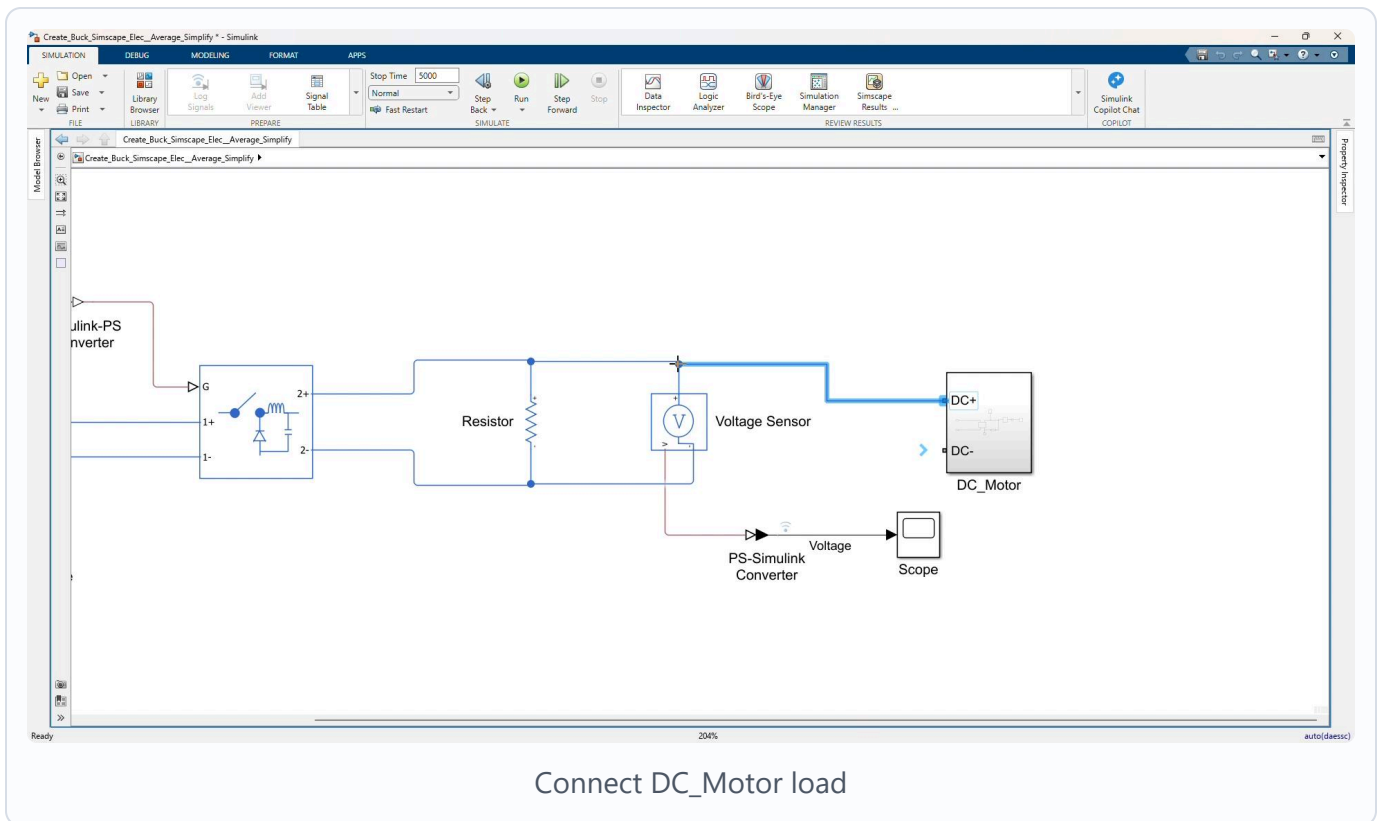
Use these mechanical load settings:

Block	Parameter	Value
Inertia	Inertia	0.01 kg*m^2
Rotational Damper	Damping	1e-3 N*m*s/rad

Connect the motor mechanical side:

DC Motor R -> Inertia -> Ideal Rotational Motion Sensor -> Rotational Damper
DC Motor C -> Mechanical Rotational Reference
Rotational Damper other side -> Mechanical Rotational Reference
Motion Sensor speed output -> PS-Simulink Converter1 -> Scope1

10. Connect the DC_Motor subsystem to the converter output



Video screenshot: DC_Motor subsystem connected in parallel with the resistor load.

Return to the top-level model.

Connect the motor subsystem in parallel with the resistor and voltage sensor:

Buck Converter p2 -> DC_Motor DC+

Buck Converter n2 -> DC_Motor DC-

Keep the voltage sensor across the output node so the voltage can still be measured.

At this point, the converter output supplies:

Resistor load + DC motor load

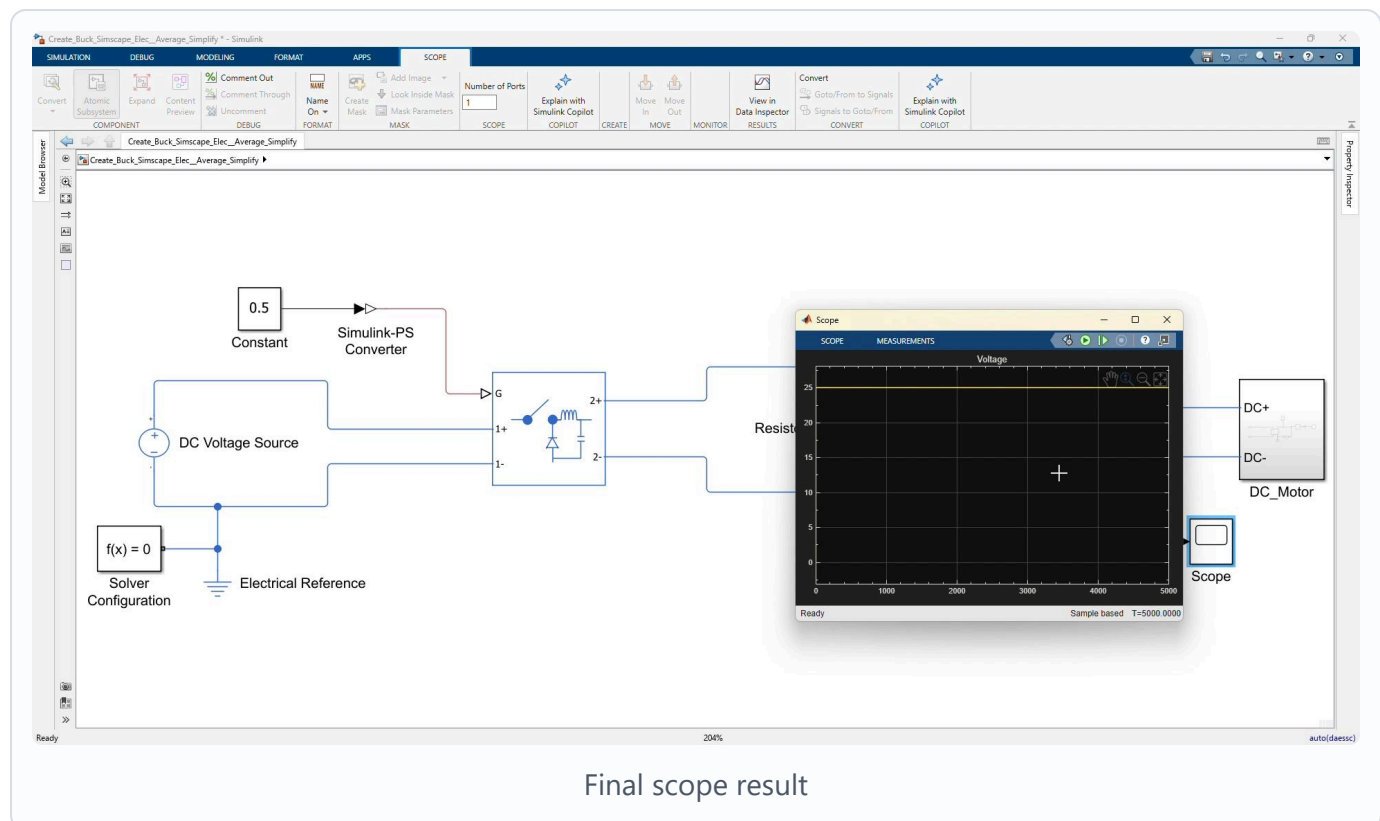
11. Increase the stop time for the motor response

Set the model stop time to:

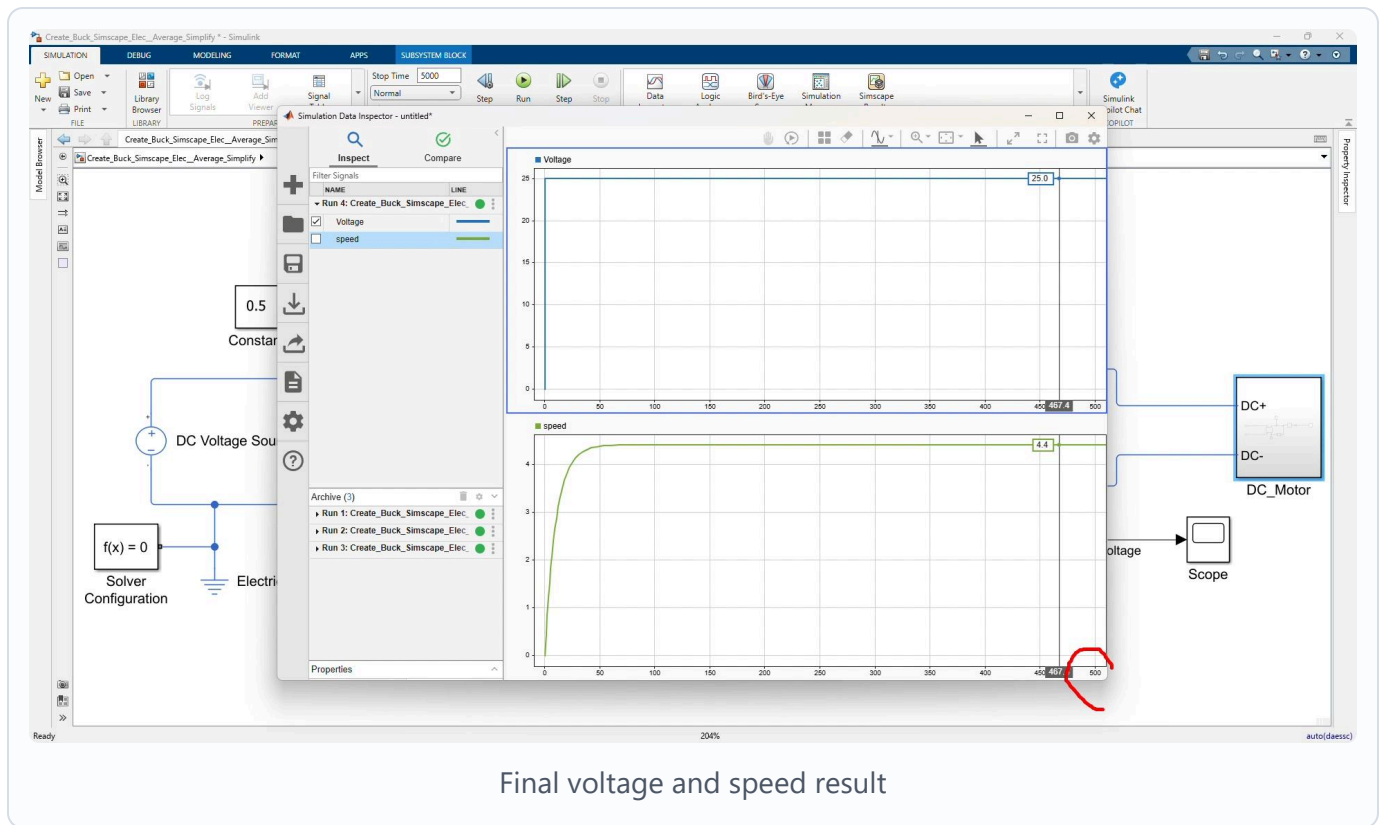
5000

The longer stop time is used because the motor mechanical speed response is slower than the electrical voltage response.

12. Run the final average buck simulation



Video screenshot: final simulation result viewed in the Scope.



Video screenshot: Simulation Data Inspector showing voltage and motor speed after the final run.

1. Run the finished model.
2. Open Simulation Data Inspector.
3. Plot the output voltage signal.
4. Plot the motor speed signal.
5. Arrange the plots so voltage and speed can be reviewed together.

Expected discussion:

- The averaged buck converter quickly produces the output voltage.
- With $V_{in} = 50 \text{ V}$ and $Duty = 0.5$, the voltage is around 25 V .
- The motor speed rises more slowly because of inertia and damping.
- The average converter model is useful for fast system-level studies where switching ripple is not the focus.

Suggested Instructor Flow

1. Open `Create_Buck_Simscape_Elec__Average_Simplify.slx`.
2. Review the prepared source, resistor load, duty command, sensor, and scope blocks.
3. Add the `Buck Converter` block.
4. Configure the converter as an averaged switching device.
5. Connect the DC source to `p1` and `n1`.
6. Connect `Constant(0.5)` through `Simulink-PS Converter` to the `G` port.
7. Connect `p2` and `n2` to the resistor load.
8. Connect the voltage sensor and scope.
9. Run and confirm about `25 V` output.
10. Create or open the `DC_Motor` subsystem.
11. Configure the DC motor, inertia, damper, speed sensor, and speed scope.
12. Connect `DC_Motor` in parallel with the resistor at the converter output.
13. Change stop time to `5000`.
14. Run the final simulation.
15. Compare output voltage and motor speed in Simulation Data Inspector.

Participant Exercise

Ask participants to record:

Item	Expected Observation
Input voltage	<code>50 V</code>
Duty command	<code>0.5</code>
Approximate no-motor output voltage	About <code>25 V</code>
Output load after final connection	Resistor and DC motor in parallel
Motor speed behavior	Gradual rise due to inertia and damping

Optional exercise:

1. Change the duty command to another value such as `0.3` or `0.8`.
2. Run the model again.

3. Observe how the output voltage and motor speed change.

Troubleshooting

Symptom	Likely Cause	Fix
Converter output is zero	Duty input is not connected to <code>G</code>	Connect <code>Constant -> Simulink-PS Converter -> G</code>
Simscape error about missing reference	Electrical reference or solver configuration is not connected	Connect <code>Electrical Reference</code> and <code>Solver Configuration</code> to the electrical network
Voltage is not shown in Scope	Voltage sensor output is not connected through <code>PS-Simulink Converter</code>	Connect <code>Voltage Sensor V -> PS-Simulink Converter -> Scope</code>
Output voltage is not near expected average value	Duty value, source voltage, or converter parameters are different	Check <code>Vin = 50 V</code> , <code>Duty = 0.5</code> , and Buck Converter settings
Motor speed does not plot	Speed sensor or <code>PS-Simulink Converter1</code> is not connected	Check <code>Ideal Rotational Motion Sensor -> PS-Simulink Converter1 -> Scope1</code>
Simulation too short for motor speed response	Stop time is still <code>500</code>	Set stop time to <code>5000</code> for the final motor-load run

Reference Check

Top-level finished model should use this structure:

Constant(0.5) -> Simulink-PS Converter -> Buck Converter G
DC Voltage Source -> Buck Converter input p1/n1
Buck Converter output p2/n2 -> Resistor
Buck Converter output p2/n2 -> Voltage Sensor -> PS-Simulink Converter ->
Scope
Buck Converter output p2/n2 -> DC_Motor DC+/DC-

DC_Motor subsystem should use this structure:

DC+ / DC- -> DC Motor electrical ports
DC Motor shaft -> Inertia and Rotational Damper
Mechanical side -> Mechanical Rotational Reference
Ideal Rotational Motion Sensor -> PS-Simulink Converter1 -> Scope1

Key average-switch concept:

The average Buck Converter model uses duty cycle directly.
It is faster for system-level simulation because it does not simulate every
PWM switching event.